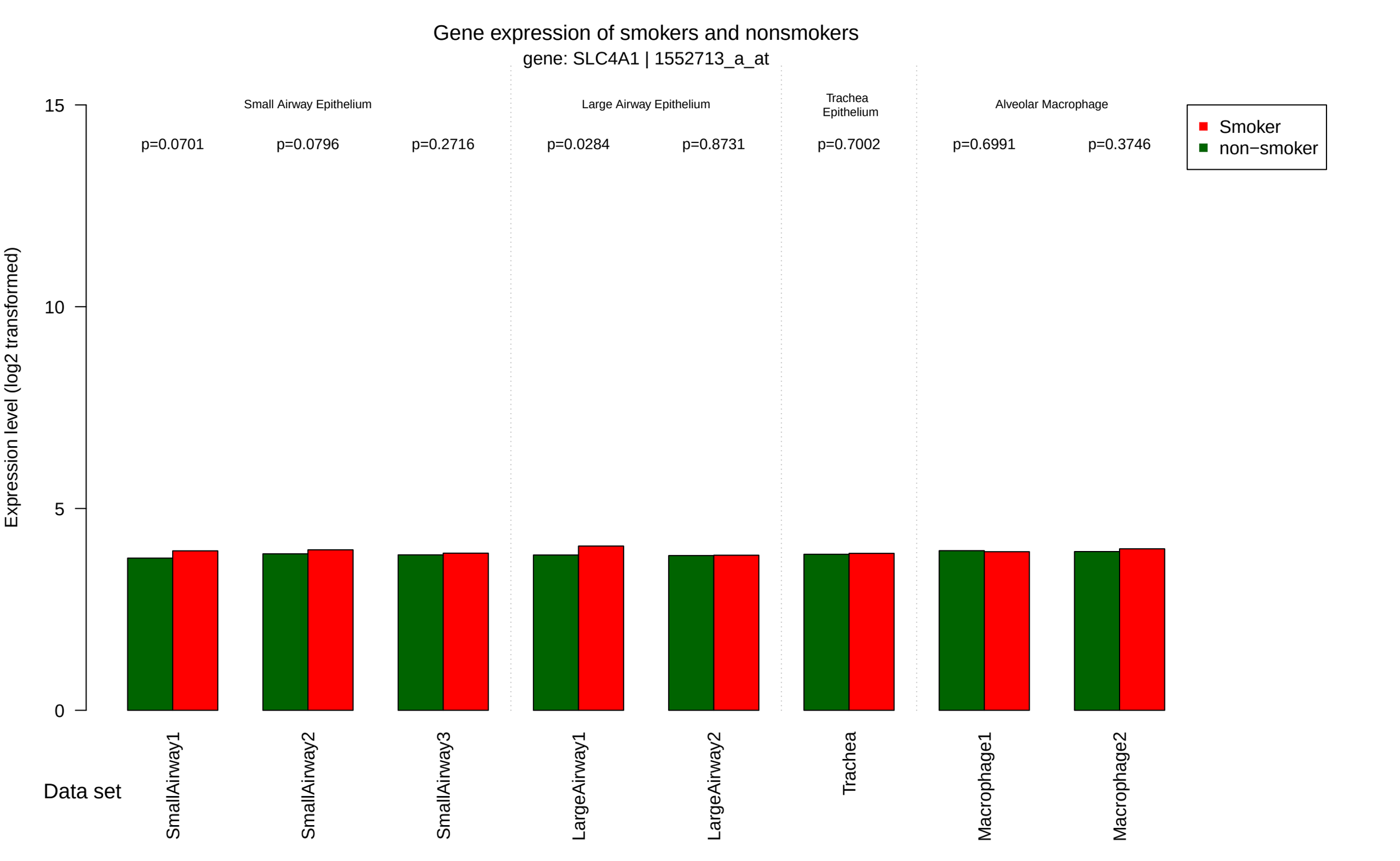
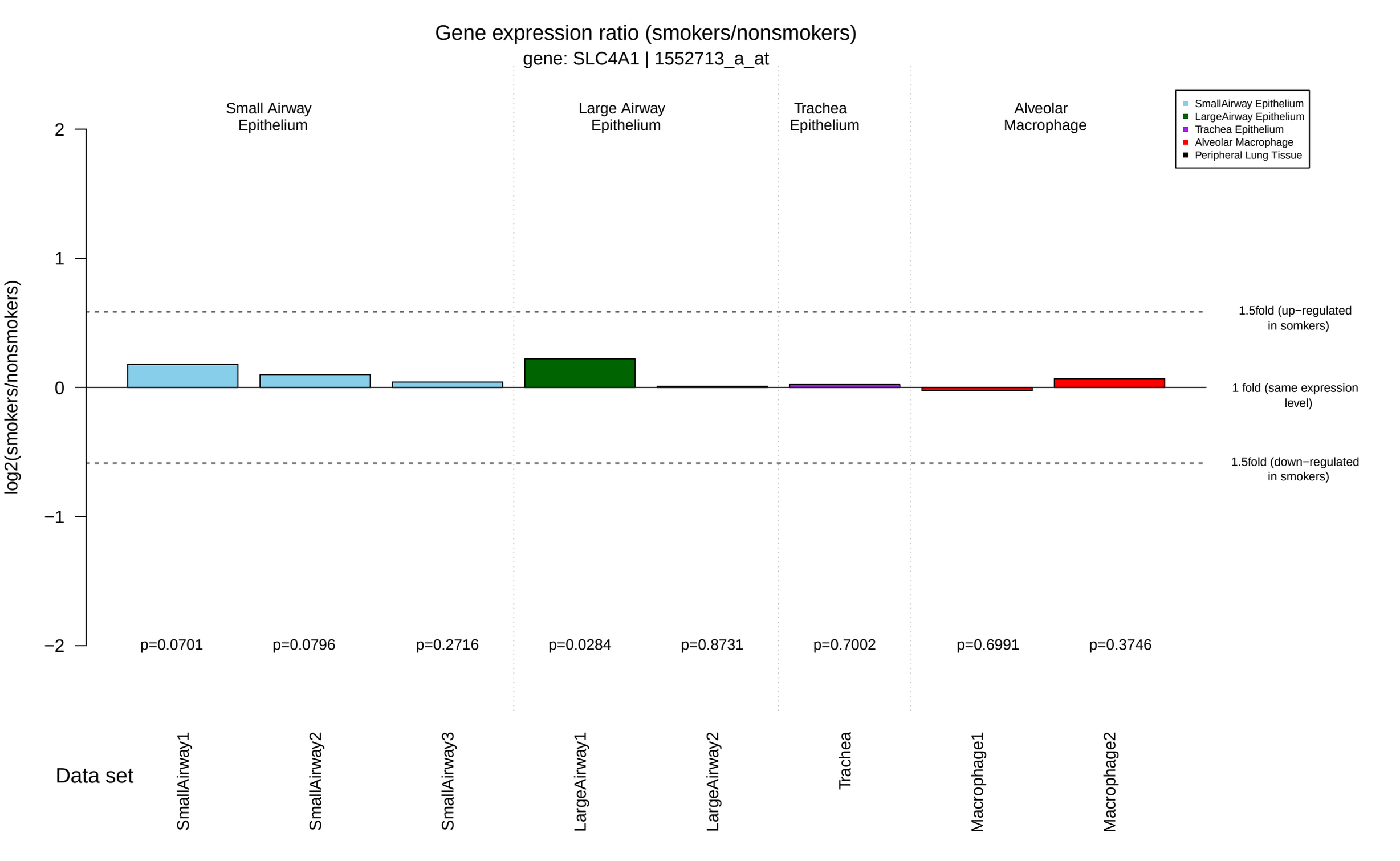
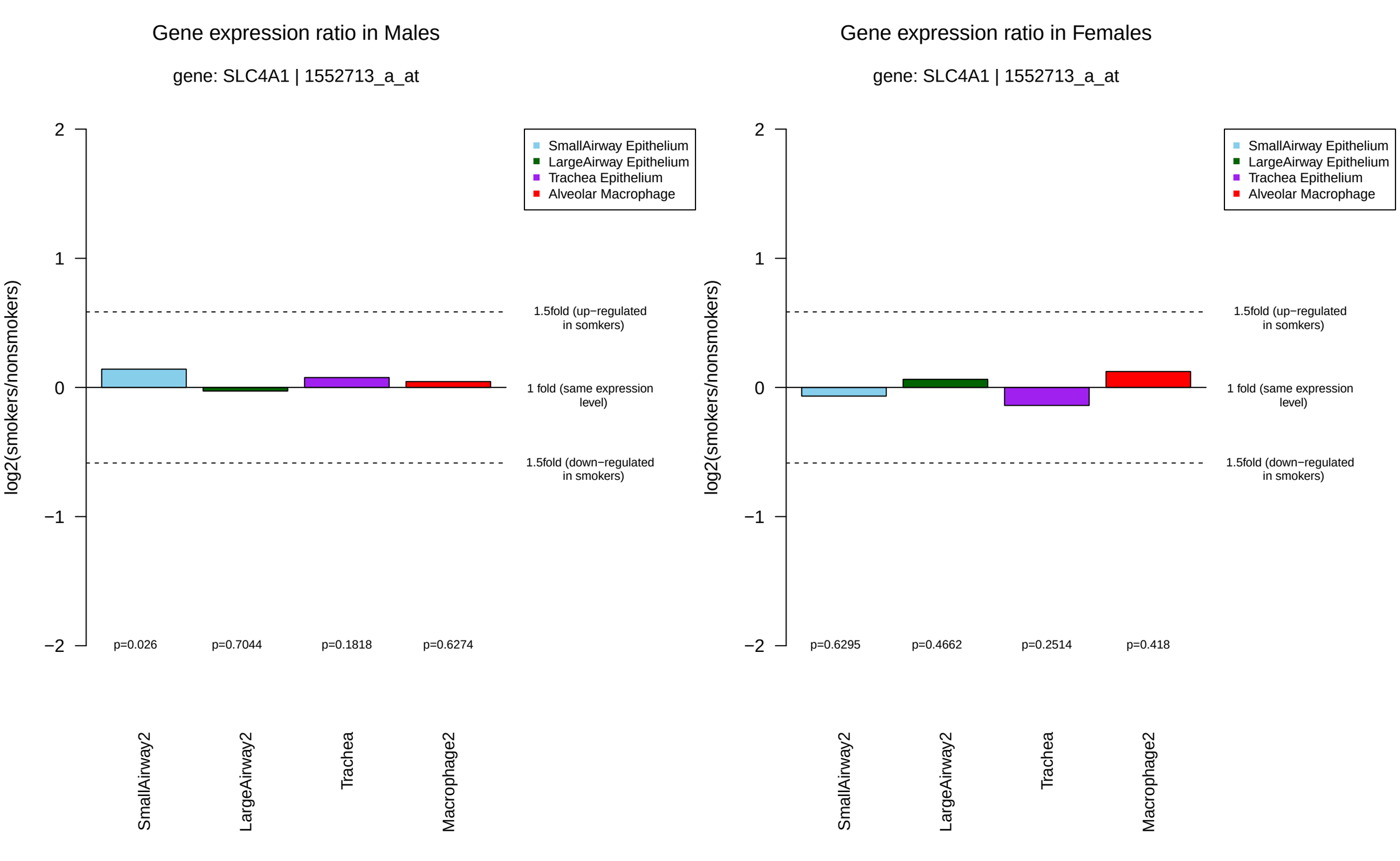
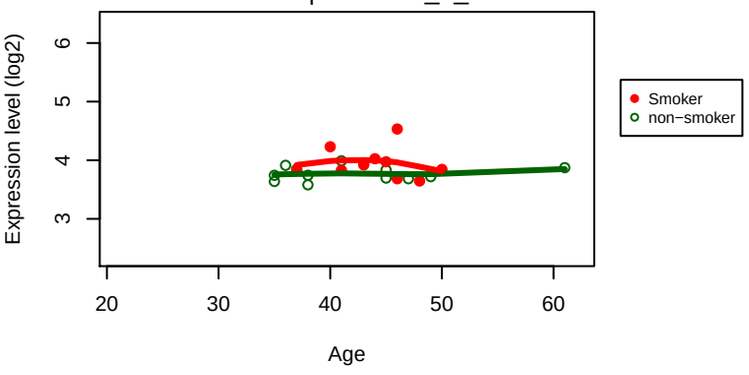
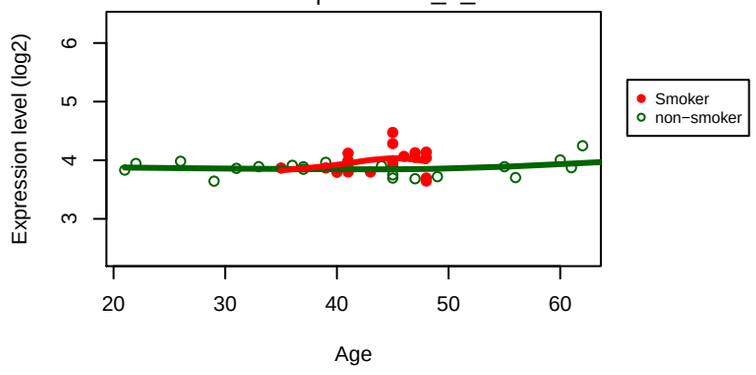
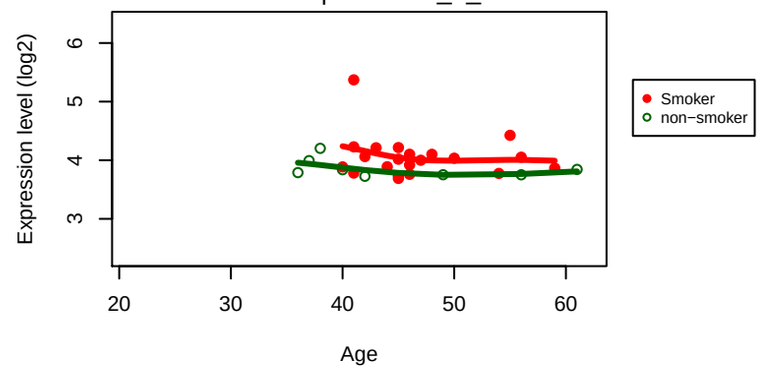

Gene expression of smokers and nonsmokers
gene: SLC4A1 | 1552713_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0701
p=0.0796
p=0.2716
p=0.0284
p=0.8731
p=0.7002
p=0.6991
p=0.3746
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

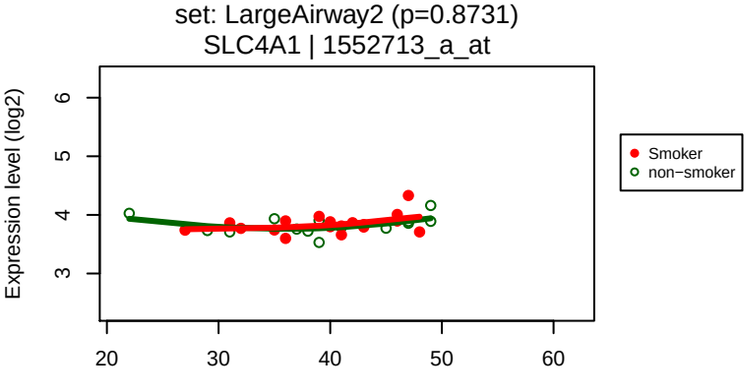

Gene expression ratio (smokers/nonsmokers)
gene: SLC4A1 | 1552713_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0701
p=0.0796
p=0.2716
p=0.0284
p=0.8731
p=0.7002
p=0.6991
p=0.3746
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

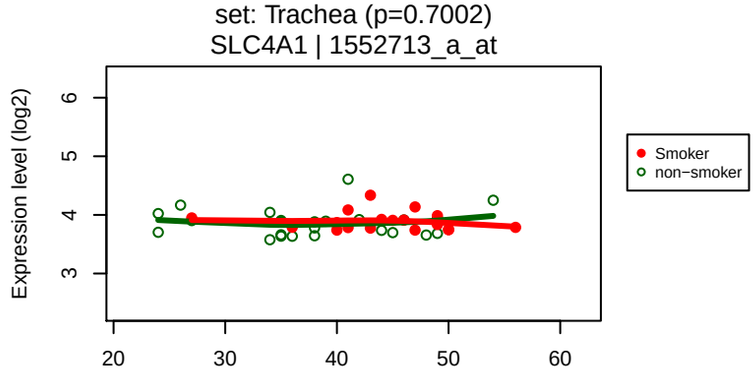

Gene expression ratio in Males
gene: SLC4A1 | 1552713_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.026
p=0.7044
p=0.1818
p=0.6274
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC4A1 | 1552713_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6295
p=0.4662
p=0.2514
p=0.418
SmallAirway2
LargeAirway2
Trachea
Macrophage2

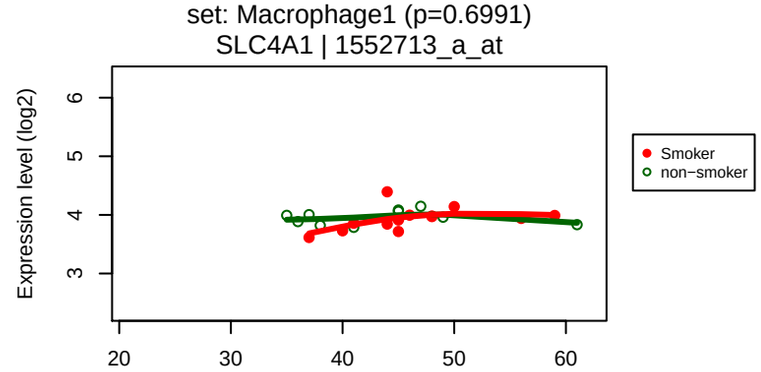

set: SmallAirway1 (p=0.0701)
SLC4A1 | 1552713_a_at
Expression level (log2)
Age
Smoker
non−smoker

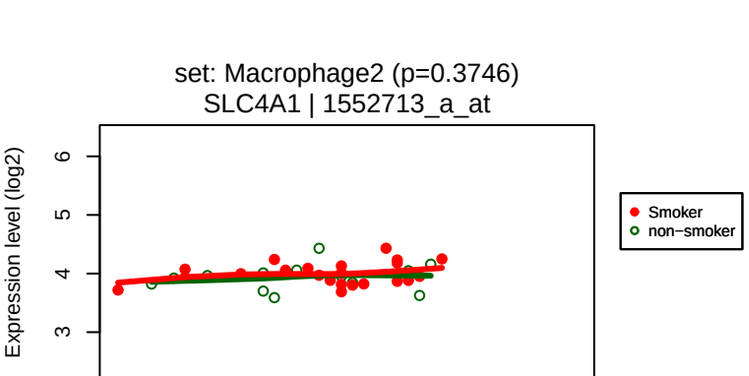

set: SmallAirway2 (p=0.0796)
SLC4A1 | 1552713_a_at
Expression level (log2)
Age
Smoker
non−smoker

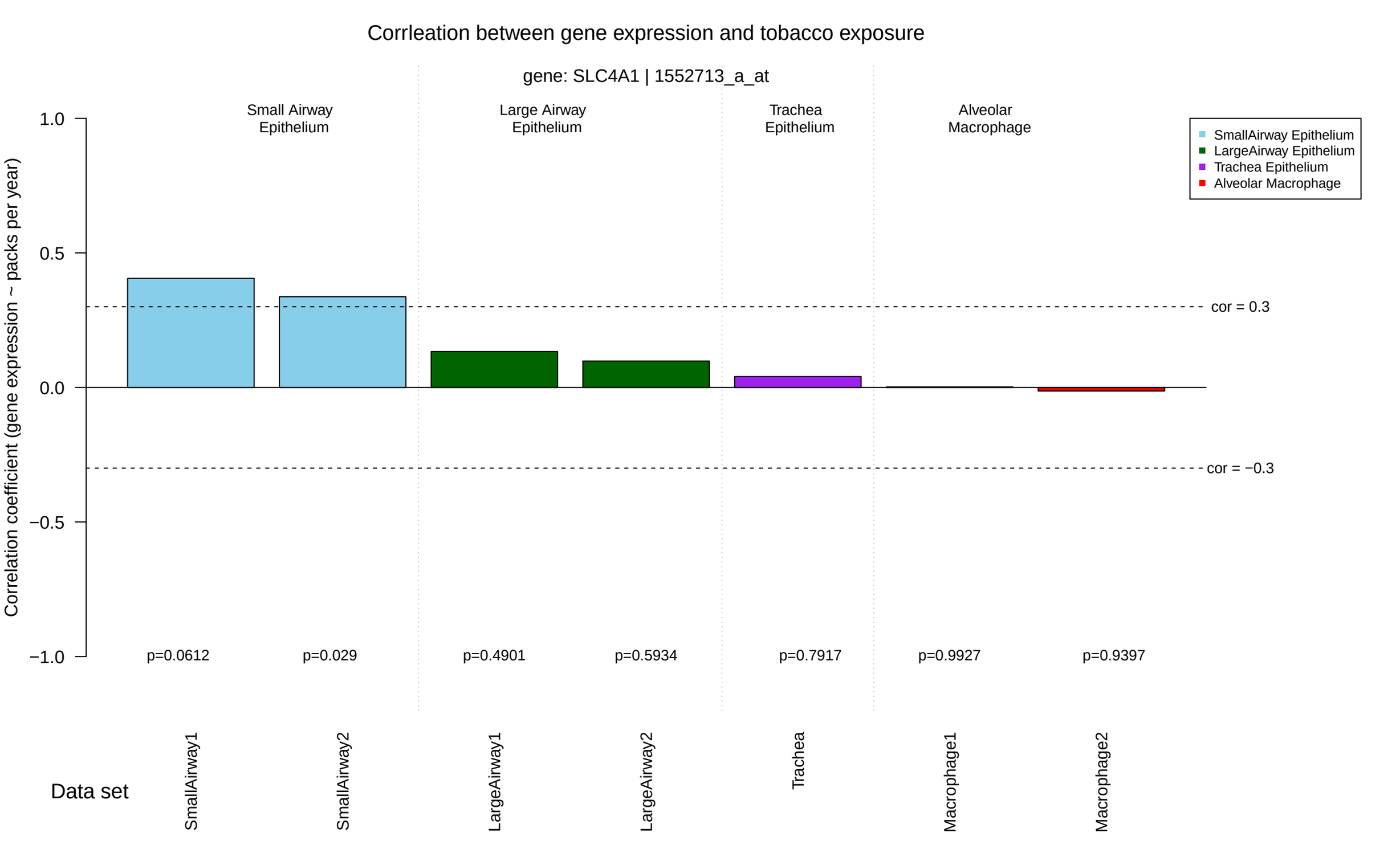

set: LargeAirway1 (p=0.0284)
SLC4A1 | 1552713_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8731)
SLC4A1 | 1552713_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7002)
SLC4A1 | 1552713_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6991)
SLC4A1 | 1552713_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3746)
SLC4A1 | 1552713_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC4A1 | 1552713_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0612
p=0.029
p=0.4901
p=0.5934
p=0.7917
p=0.9927
p=0.9397
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2